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Advancing the concept of technical ability of force application in sprint acceleration

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ABSTRACT

Technical ability of force application (sagittal force vector orientation) is considered a performance determinant in sprint acceleration, yet the underlying kinematics require further investigation. Sixteen elite male bobsleigh athletes performed unresisted and resisted (85 kg sled push) accelerations initiated with full effort, while ground reaction forces and kinematics were recorded for ground contacts 1 to 5. Ratio of forces (RF), body position, centre of mass velocity components, and lower limb segment rotations were examined for correlations and differences between acceleration tasks. RF was significantly associated with body position (unresisted: $r = .82$, $p < .001$; resisted: $r = .83$, $p < .001$) and extensional velocity components (unresisted: $r = .83$, $p < .001$; resisted: $r = .90$, $p < .001$). There was also a significant correlation between body position and extensional velocity component (unresisted: $r = .98$, $p < .001$; resisted: $r = .84$, $p < .001$) regardless of acceleration task. Shank orientation significantly explained variance in body position (unresisted: adj. $R^2 = .86$, $p < .001$; resisted: adj. $R^2 = .81$, $p < .001$). Resisted acceleration provoked greater RF and forward orientations as well as higher shares of extensional velocity components. To improve technical ability, practitioners should focus on manipulating body position to increase the extensional velocity component as well as on shank segment rotation in preparation for the ground contact phase.

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Introduction

Increasing forward velocity of the body's centre of mass (CoM) during sprinting is achieved through the application of a forward-directed net impulse of the ground reaction force (GRF). This motor skill is crucial for sport performance in various team sports (e.g., rugby, football, soccer) as well as in athletics and winter sport (e.g., 100 m-sprint, long jump, bobsleigh, skeleton). To design experimental studies and

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performance enhancement programmes a thorough comprehension of performance-associated factors of acceleration is required.

It is evident that strong and powerful lower limb muscles are indispensable to create the GRF needed to propel the body forward. Numerous studies on accelerated running have identified the ankle plantar flexors and hip extensors as the primary sources of propulsion (Morin et al., 2015; Pandy et al., 2021; Schache et al., 2019), with the hip extensors being especially important for resisted acceleration tasks such as the push phase in bobsleigh (Zedler et al., 2025).

Besides the ability to generate high GRF, the technical ability to direct the vector of the GRF into the intended direction of motion is crucial, which is referred to as the ratio of anterior to resultant GRF (RF) or mechanical effectiveness (Morin et al., 2011). It is even argued that orientation of the GRF vector is more important than its magnitude (Kawamori et al., 2013; Kugler & Janshen, 2010; Morin et al., 2011). Although manipulating GRF vector orientation is considered a technical skill, the underlying kinematics remain poorly understood. Previous studies on unresisted acceleration have demonstrated that GRF orientation is strongly related to the position of the CoM relative to the point of force application (Bezodis et al., 2015, 2017; King et al., 2022; Kugler & Janshen, 2010). Findings on resisted acceleration exercises such as the sled-tow or sled-push have revealed that accelerating an external mass acutely increases forward orientation of the body (Lockie et al., 2003) and improves the technical ability of force application (Pantoja et al., 2018). Training interventions involving external resistance have also shown that this effect is transferable to unresisted acceleration (Morin et al., 2017). However, the underlying kinematic relationships are limited to spatio-temporal variables (Lockie et al., 2003; Pantoja et al., 2018; Pareja-Blanco et al., 2022) joint angles (Osterwald et al., 2021) and trunk orientation (Lockie et al., 2003). By contrasting resisted and unresisted acceleration tasks, kinematic features underpinning the effective GRF orientation can be identified.

On a whole-body level, acceleration results from a coordinated combination of CoM rotation and CoM extension (Figure 1).

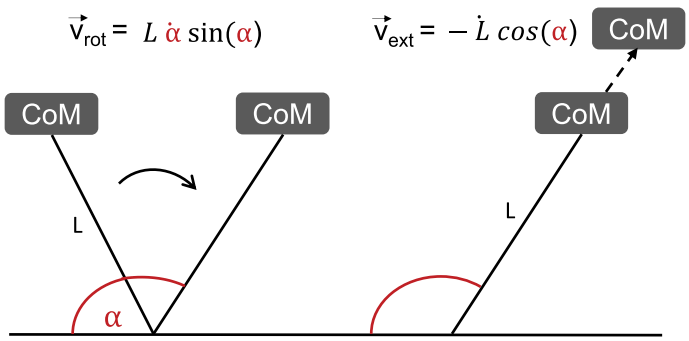

Figure 1. Simple model of a body's centre of mass (CoM) mounted on a leg segment (L) with corresponding CoM-angle α defined as the sagittal angle between L and the ground. Rotational v_{rot} (left) and extensional v_{ext} (right) components of forward centre of mass velocity according to Jacobs and van Ingen Schenau (1992).

During ground contact, the CoM rotates about the point of force application and extends by increasing its distance from the point of force application, thereby influencing the orientation and magnitude of the resultant GRF (Jacobs & van Ingen Schenau, 1992). The corresponding rotational and extensional velocity components quantify how much each movement strategy—rotation or extension—contributes to forward velocity. Since both components depend on the CoM-angle α (Figure 1), body position directly influences how effectively rotation and extension translate into forward velocity, which we term propulsive effectiveness henceforth. The propulsive effectiveness is defined as $sin(\alpha)$ for CoM rotation and $-1 \cdot cos(\alpha)$ for CoM extension (Figure 2). For resisted acceleration greater forward body positions (i.e., greater CoM-angle) compared to unresisted can be expected, which corresponds to a right-shift of the sector covered by the CoM-angle during stance. Consequently, altered body position affects the propulsive effectiveness of rotation and extension. For instance, shifting the sector of the CoM-angle from 85–130° to 95–140° would decrease the stance-averaged propulsive effectiveness of rotation by 7% and increase the propulsive effectiveness of extension by 54%. Whether this increase will shift the kinematic strategy to accelerate from rotation to extension remains to be proven.

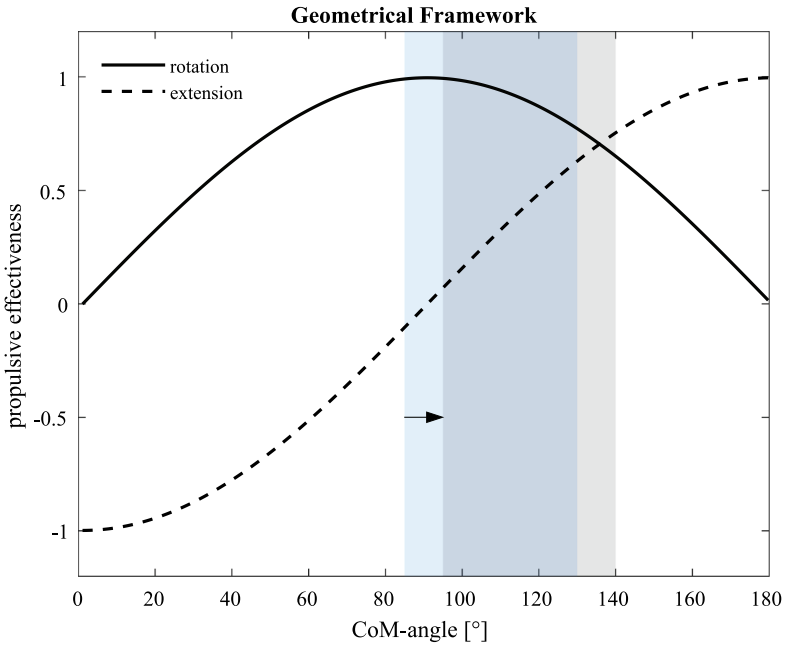

Figure 2. Propulsive effectiveness of centre of mass (CoM) rotation and extension to increase forward velocity of the centre of mass expressed as the *sine* and *-1·cos(α)* of the CoM-angle (sagittal angle between the ground and the vector pointing from point of force application to the CoM). Angles of 90° represent foot placements directly under the CoM, while angles greater than 90° correspond to posterior foot placement (forward lean). The shaded area in blue resembles a hypothetical ground contact phase during sprint acceleration. A greater forward-oriented CoM-angle (grey area) during the ground contact phase directly affects the effectiveness to convert CoM rotation or extension into forward velocity. This geometrical framework can be applied to individual lower-limb segment analyses for propulsive effectiveness of rotation.

Due to the linked-segment nature of the musculoskeletal system, body position as well as CoM rotation and extension are a result of body segment configuration and rotations, respectively. To tailor fast feedback technical interventions, it is therefore necessary to understand how differences in body position are realised at a segment level and how coordinated segmental rotations allow for effective GRF application to increase forward velocity. It was suggested that body position primarily depends on stance limb configuration (Bezodis et al., 2017), with the shank identified as a key segment in setting up a body configuration conducive to horizontal force production (Alt et al., 2022; King et al., 2022). The geometrical framework introduced to examine CoM kinematics (Figure 2) can be further extended to investigate the contribution of segmental rotations to propulsion, given that rigid body segments can undergo rotation but not extension.

Consequently, the purpose of this study was to advance the concept of technical ability in sprint acceleration by examining the underlying characteristics of body position and CoM velocity components, as well as lower-limb segment rotations. By contrasting resisted and unresisted tasks, we aimed to determine how kinematic adaptations influence the effectiveness of force application during the initial acceleration phase (ground contacts 1 to 5). We hypothesised that (1) the RF is positively correlated with the CoM-angle at initial ground contact (angle of attack) and at toe-off (take-off angle), and the CoM extensional velocity component, (2) resisted acceleration provokes greater angles of attack and take-off angles, shifting the kinematic strategy to accelerate towards CoM extension, and (3) the angle of attack is primarily determined by stance limb configuration with the shank exerting the greatest influence.

Materials and methods

Participants

Sixteen elite male bobsleigh athletes participated in the study (body height: 1.88 ± 0.06 m, body mass: 100.8 ± 6.3 kg, age: 26.3 ± 4.3 years). All athletes belonged to the German Olympic bobsleigh squad and provided informed consent prior to participation in the data collection. The procedures of the study were approved by the university's internal committee for the protection of human participants in accordance with the Declaration of Helsinki (ethics number: 67/2020).

Study design

In this cross-sectional study, the athletes performed both resisted and unresisted acceleration tasks with maximum effort, while ground reaction forces and kinematics of the athletes were simultaneously recorded with opto-electronic motion capture. For resisted acceleration, the athletes pushed a custom-made bobsled dummy mounted on wheels along guiding rails. To ensure a high practical relevance for bobsleigh, the mass of the sled dummy (85 kg) was chosen to approximate the mass which must be accelerated by one athlete in a two-man bobsleigh competition. For more detailed specifications regarding the sled dummy see Zedler et al. (2025).

Experimental protocol

First, body segment lengths and circumferences were measured, and anatomical landmarks for marker placement were palpated and marked on the skin by pen. Afterwards, the athletes performed their individual warm-up to prepare for maximal effort sprint acceleration. The markers were firmly attached to the skin of the athletes after the warm-up, just before the main measurements. The athletes then performed three unresisted sprints with at least 5 min rest in between to ensure maximum effort levels for each run. The athletes initiated the unresisted sprints from a 3-point crouch start with one hand touching the athletics track and accelerated for 15 m. After the unresisted sprints, the athletes did three resisted sprints by pushing a bobsled dummy (mass: 85 kg), initiating the start from a brakeman position with both hands holding the sled handles. The order of unresisted followed by resisted sprints was kept constant to avoid potential carry-over effects of resisted sprints on subsequent unresisted running mechanics. Bobsleigh athletes commonly perform unresisted sprints as part of their warm-up routines before practicing sled pushes. Therefore, unresisted sprints were not expected to meaningfully affect subsequent resisted sprint performance. Sufficient rest periods were provided between trials and conditions to mitigate potential fatigue effects.

Data collection & processing

Ground reaction forces were collected at 1250 Hz using 13 digital force plates embedded into the athletics track in series (Model: 9667AA, 600 × 1200 mm, Kistler Instrumente AG, Winterthur, CH). The positional data of retroreflective markers (diameter: 14 mm) were synchronously recorded by 25 infrared cameras (Arqus A5, Qualysis AB, Gothenburg, SW) at 250 Hz. Both force and marker data were filtered using a recursive low-pass Butterworth filter at a cut-off frequency of 25 Hz. For each experimental condition and athlete, the fastest run based on average acceleration from start until the 5 m-mark was selected for further analyses. This approach was chosen to characterise kinematic strategies associated with effective force application under maximal effort. Five consecutive ground contact phases after the initial start were extracted by applying a threshold of 20 N to the raw vertical GRF. The body's CoM of the athletes was estimated based on De Leva (1996). The following dependent variables were derived from the dataset for statistical analysis (Table 1).

Statistics

The relationship between mechanical effectiveness and kinematic variables (Table 1) was examined using Pearson product-moment correlation coefficients, separately for unresisted and resisted acceleration tasks. For each athlete and task, the ratio of forces, extensional contribution, angle of attack and take-off angle were averaged across five consecutive ground contact phases to obtain step-averaged values representative of early acceleration mechanics. The level of significance was set at $p < 0.05$.

The effect of external resistance on the dependent variables (Table 1) was investigated using a repeated-measures analysis of variance with two factors (acceleration task: unresisted, resisted; step number: ground contact phases 1–5).

Table 1. Definition of dependent variables.

Variable	Unit	Definition
Ratio of forces (RF)	dimensionless	Ratio of average antero-posterior to the average resultant of antero-posterior and vertical ground reaction force according to Bezodis et al. (2021).
CoM-angle	°	Centre of mass angle defined as the sagittal angle between the ground and the directional vector pointing from centre of force application under the foot to the body's centre of mass.
Angle of attack	°	CoM-angle at the instant of initial ground contact (TD).
Take-off angle	°	CoM-angle at the instant of take-off (TO).
CoM rotation	°	Change in CoM-angle during a ground contact phase.
CoM extension	m	Change in distance between the centre of force application under foot and the body's centre of mass.
Rotational component	ms^{-1}	Velocity component of forward centre of mass velocity induced by CoM rotation (Jacobs & van Ingen Schenau, 1992).
Extensional component	ms^{-1}	Velocity component of forward centre of mass velocity induced by CoM extension (Jacobs & van Ingen Schenau, 1992).
Extensional contribution	%	Ratio of extensional velocity component and total forward CoM velocity expressed as a percentage.
Contact range of motion	°	Change in CoM-angle from initial ground contact until take-off.
Propulsive effectiveness	dimensionless	Potential to convert either rotation or extension to forward centre of mass velocity expressed as the sine (rotation) or negative cosine (extension) of the CoM-angle (or segment angle when referring to a specific lower limb segment).

Assumptions of normality and sphericity were verified prior to repeated-measures ANOVA. In case of violated sphericity (Mauchly's test), Greenhouse-Geisser corrections were applied. To further explain task-specific differences in the rotational and extensional components of forward CoM velocity, a mechanistic log-difference decomposition was applied to the analytical expressions describing these components (Figure 1). This analysis was done retrospectively as an explanatory tool to quantify the relative contribution of underlying kinematic factors to observed between-task differences. For each participant and acceleration task, step-averaged values of all variables were first computed separately for each of the five analysed ground contact phases. Subsequently, values were averaged across the five ground contacts to obtain a single representative value per participant and task. Relative changes between resisted and unresisted acceleration were then calculated as logarithmic differences. To quantify the relative contribution of each term to the observed change in the respective velocity component, the absolute magnitude of each Δlog term was expressed as a proportion of the summed absolute Δlog terms within each decomposition. These proportional contributions were subsequently averaged across participants and reported as percentages.

Time-series of velocity components and segmental rotations were analysed using statistical parametric mapping (SPM) according to (Pataky, 2010). A multiple linear regression analysis was conducted to examine the relationship between the angle of attack and the segment angles of the stance limb (thigh, shank, and foot). Assumptions of linearity, independence, homoscedasticity, and normality of residuals were checked before analysis. The statistical significance of each predictor was assessed using standardised regression coefficients (β) and p-values, with the significance set at $p < 0.05$. The overall model fit was evaluated using the coefficient of determination (R^2) and adjusted R^2. All calculations were conducted in Matlab (MATLAB 2021a, The Mathworks Inc., Natick, MA, USA).

Results

Ratio of forces and extensional contribution were significantly associated with both the angle of attack and take-off angle during resisted acceleration, and with the take-off angle only during unresisted acceleration (Table 2). In both tasks, greater forward orientation of the associated angles corresponded to higher RF and greater extensional contribution. In addition, RF and extensional contribution were significantly and positively associated irrespective of acceleration task. All mean values of all variables in Table 2 were significantly greater for the resisted acceleration task compared to unresisted.

Significant task and step effects with large effect sizes were found for the angle of attack, take-off angle, RF, and extensional contribution (Table 3). These variables were significantly higher for resisted acceleration compared to unresisted and consistently decreased with increasing step number. Task × step interaction effects indicated greater task differences with increasing step number for take-off angle, RF and extensional contribution. The sector of the CoM-angle covered during the ground contact phases (contact range of motion) increased with increasing step number as well but remained similar across acceleration tasks (Figure 3).

Waveform analysis of the rotational and extensional components of forward CoM velocity revealed significant main and interaction effects for the rotational and extensional components as well as for the extensional contribution. Supra-threshold clusters resulting from post-hoc analysis are marked in Figure 4. For each ground contact phase analysed a significant task effect was found for rotational and extensional velocity components and for extensional contribution. Resisted acceleration provoked a decrease in rotational and an increase in the extensional velocity component, which was also reflected in a higher extensional contribution for all ground contact phases.

Since the angle of attack is an overall measure of whole-body posture, the contribution of individual body segments was analysed using multiple linear regression. The variance in the step-averaged angles of attack (ground contact 1 to 5) could be significantly explained by sagittal body segment orientations at the instant of initial ground contact for both resisted (adj. $R^2 = .81$, $p < .001$) and unresisted (adj. $R^2 = .86$, $p < .001$) acceleration tasks (Table 4). Based on standardised coefficient estimates, the shank segment angle exerted the greatest explanatory power regardless of acceleration task.

Further analysing the progression of segment rotations throughout the ground contact phases provided valuable insights into the kinematic strategy to produce propulsion on a segment level (Figure 5). Therefore, a two-way repeated-measures SPM ANOVA

Table 2. Pearson correlation coefficients grouped by acceleration task.

Task	Variable	1	2	3	4	M	SD
unresisted	1. Ratio of forces	–				.28[b]	.02
	2. Extensional contribution	.83***	–			17.2[b]	1.6
	3. Angle of attack	.46	.09	–		75.4[b]	2.3
	4. Take-off angle	.82***	.98***	.09	–	128.8[b]	2.0
resisted	1. Ratio of forces	–				.44[a]	.03
	2. Extensional contribution	.90***	–			29.5[a]	3.2
	3. Angle of attack	.83***	.75***	–		88.9[a]	5.5
	4. Take-off angle	.59*	.84***	.29	–	139.3[a]	2.1

M = mean; SD = standard deviation. ***$p < .001$; **$p < .01$; *$p < .05$.
Group means sig. different to [a]unresisted and [b]resisted.

Table 3. Mean and standard deviation of selected variables for each ground contact phase grouped by acceleration task, and two-way (task = resisted, unresisted, step = ground contact 1 to 5) repeated measures ANOVA results including effect sizes.

Variable	Task	GC1	GC2	GC3	GC4	GC5	Main effect F (partial η^2)		
							Task	Step	Task × Step
Angle of attack (°)	unresisted	91.6 ± 3.6	84.4 ± 3.8	73.6 ± 4.5	67.3 ± 3.8	60.0 ± 1.9	70.3*** (.82)	372.6*** (.96)	.40 (.03)
	resisted	103.8 ± 6.7	97.6 ± 5.9	87.6 ± 7.1	81.2 ± 5.9	74.1 ± 6.7			
Take-off angle (°)	unresisted	135.6 ± 2.4	132.7 ± 2.6	128.7 ± 1.6	126.1 ± 2.9	121.1 ± 2.4	171.3*** (.92)	314.6*** (.95)†	22.2*** (.60)
	resisted	143.3 ± 1.85	141.9 ± 2.1	139.6 ± 2.5	137.3 ± 2.5	134.7 ± 2.9			
Contact range of motion (°)	unresisted	43.9 ± 4.6	48.3 ± 3.5	55.1 ± 5.2	58.8 ± 3.5	61.4 ± 3.1	3.0 (.17)	152.8*** (.91)†	1.7 (.10)†
	resisted	39.5 ± 6.6	44.1 ± 5.8	51.9 ± 6.7	56.1 ± 6.2	60.6 ± 6.3			
Ratio of forces	unresisted	.37 ± .04	.33 ± .04	.27 ± .03	.24 ± .03	.19 ± .02	327.9*** (.96)	172.5*** (0.92)	0.19 (.01)
	resisted	.52 ± .04	.48 ± .04	.43 ± .04	.39 ± .03	.35 ± .04			
Extensional contribution (%)	unresisted	26.5 ± 2.9	20.1 ± 2.6	14.8 ± 1.2	12.9 ± 1.8	11.7 ± 1.1	202.6*** (.93)	420.9*** (.97)	19.5*** (.57)
	resisted	40.6 ± 4.2	35.3 ± 4.2	27.9 ± 4.2	23.8 ± 3.2	20.0 ± 3.1			

Values are means ± standard deviation. †Greenhouse-Geisser correction where sphericity was violated.
***$p < .001$; **$p < .01$; *$p < .05$.

Table 4. Multiple regression to explain the angle of attack for the ground contact phases 1 to 5 (GC), based on sagittal angles of the trunk, thigh, shank and foot at the instant of initial ground contact. Segment angles were standardised to enable direct comparison between coefficient estimates.

Task	GC	Intercept	Trunk	Thigh	Shank	Foot	R^2	adj. R^2	p-value
unresisted	1	0	.25	.35	.84	−.30	.76	.67	<.001
	2	0	−.28	.51	.87	−.31	.64	.51	.02
	3	0	.05	.82	.30	−.07	.71	.61	.01
	4	0	−.08	.23	.45	−.64	.47	.27	.11
	5	0	.09	.52	.43	−.42	.40	.18	.19
	1–5	75.39	1.53	2.40	9.94	−.55	.89	.86	<.001
resisted	1	0	.12	.50	.15	.39	.89	.85	<.001
	2	0	−.07	.22	.76	.06	.77	.68	<.001
	3	0	.15	.38	.84	−.28	.94	.91	<.001
	4	0	.29	.14	.66	−.20	.64	.51	.02
	5	0	.14	.43	.61	−.12	.87	.82	<.001
	1–5	88.29	.64	2.03	10.74	−.86	.86	.81	<.001

(task × step) was conducted across the time-normalised lower limb segment angles. There were significant main effects of task and step for the thigh, shank and foot segment angles. No significant task × step interactions were found, indicating that the task effect was consistent across all steps. The foot and shank segments were significantly more forward oriented for resisted acceleration regardless of step number with supra-threshold clusters found for the entire ground contact phase [$p < .001$, 0–100%]. Orientation of the thigh segment was similar between the acceleration tasks from the initial ground contact until 72% of the ground contact phase and diverged thereafter with significantly more forward orientation for resisted acceleration [$p = .031$, 72–100%], suggesting increased thigh forward rotation.

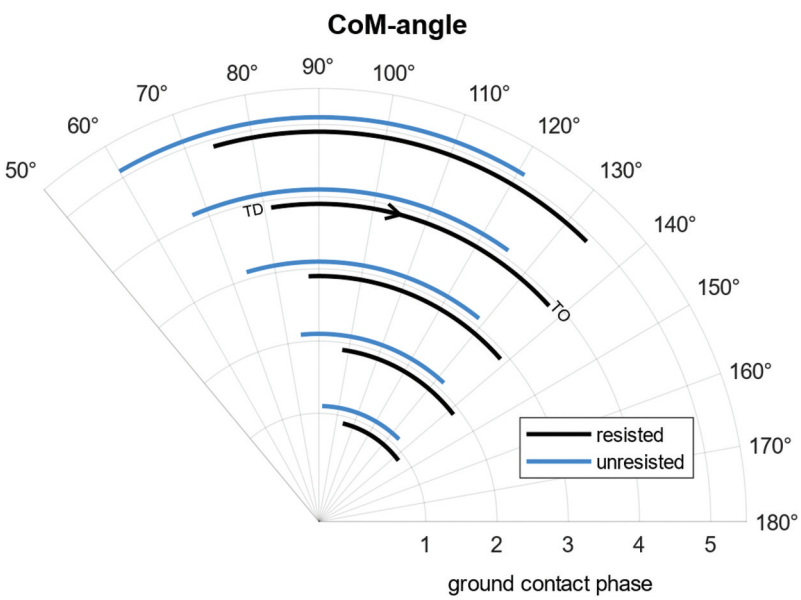

Figure 3. Angle between the ground and the vector pointing from point of force application on the ground to the body's centre of mass (CoM-angle) during the ground contact phases (1–5) from initial ground contact (TD) until toe-off (to), grouped by acceleration task (resisted = black; unresisted = blue).

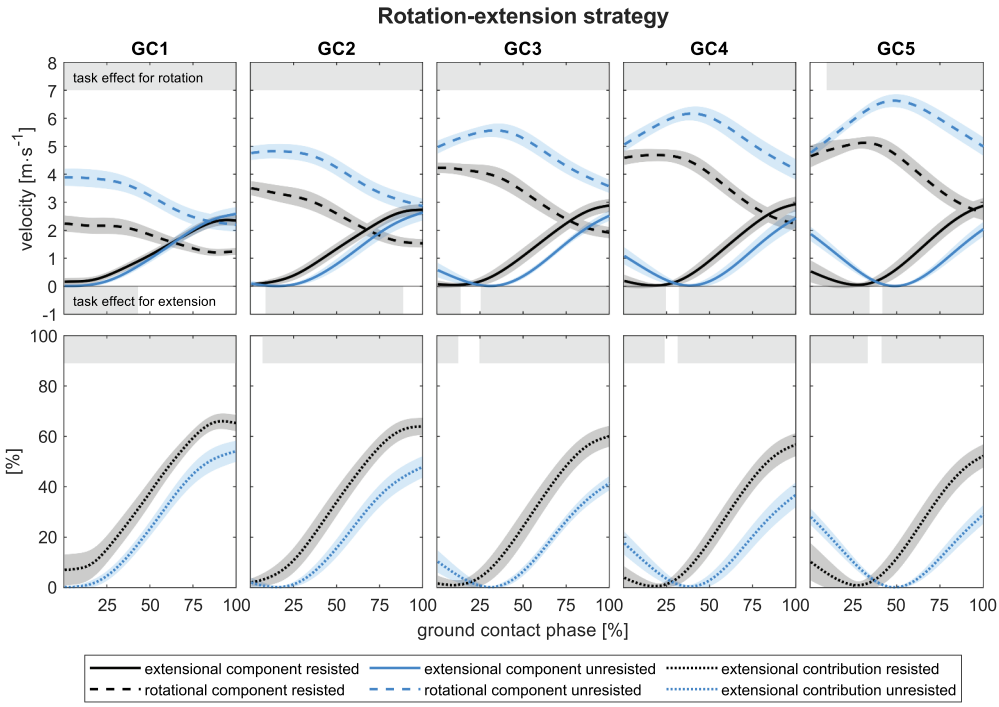

Figure 4. Rotational and extensional velocity components of forward centre of mass (CoM) velocity (top row) and extensional contribution (bottom row) expressed as the share of the extensional velocity component in forward CoM velocity. Data are time-normalised to the ground contact phases 1–5 (columns) and expressed as means and standard deviation grouped by acceleration condition (resisted = black; unresisted = blue). Statistically significant difference between the resisted and unresisted tasks are indicated by the grey horizontal boxes at the top and bottom of each subplot.

Discussion

The purpose of this study was to examine how body configuration and segmental kinematics contribute to the technical ability of force application during sprint acceleration, using resisted and unresisted tasks to manipulate mechanical demands. By linking the RF, CoM kinematics, and lower-limb segment orientations, this study aimed to advance the understanding of mechanical effectiveness in sprint acceleration.

In line with our hypotheses, the main findings were that (1) the RF was related to body position and the extensional velocity component of the CoM irrespective of acceleration task, (2) resisted acceleration provoked greater RF and increased forward orientation of the angle of attack and take-off angles as well as lower limb segment angles, and that (3) the shank emerged as a key segment in modulating the angle of attack as well as in setting up lower limb segment chain for effective horizontal force production.

The RF observed in this study are in accordance with previous studies on unresisted and resisted acceleration (King et al., 2022; Zedler et al., 2025). RF is mathematically strongly related to the sagittal GRF vector orientation. Therefore, the significant association of RF with CoM-angle found in this study supports the close relationship between GRF orientation and body position (Kawamori et al., 2013; Kugler & Janshen, 2010; Morin et al., 2011) and further highlights task-

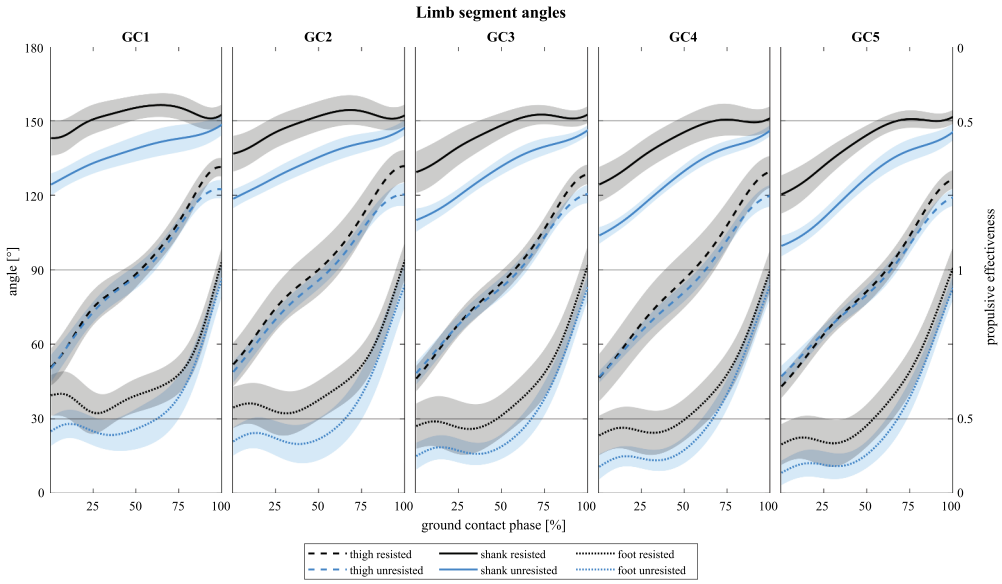

Figure 5. Sagittal orientation of the thigh, shank and foot segment during the ground contact phases 1–5 (GC1-5), displayed as group averages including standard deviation (shaded area). The propulsive effectiveness (sine of the segment angle) is shown on the right vertical axis.

specific peculiarities. In resisted acceleration, RF was both associated with the angle of attack (CoM-angle at TD) and the take-off angle (CoM-angle at TO), whereas in unresisted acceleration RF was only associated with the take-off angle. A more forward-oriented body position during stance can be attained by increasing either the angle of attack, the take-off angle, or both. The present findings indicate that, during unresisted acceleration, changes in RF are more sensitive to variations in the take-off angle than to the angle of attack, whereas in the resisted task the opposite pattern was observed, with RF being more sensitive to changes in the angle of attack. Similarly, Kugler and Janshen (2010) have shown that increased take-off angles during unresisted acceleration were also significantly related to GRF vector forward orientation at the instant of maximum GRF production. However, increasing take-off angles challenge the ability to maintain balance. Elite sprinters achieve this by rapidly retracting the swing leg, which counteracts the forward angular momentum of the body's CoM (Moir et al., 2018). In resisted acceleration, balance control is less critical, as athletes continuously stabilise themselves by holding the sled handles. Beyond confirming the relationship between GRF vector and whole-body orientation, this study further demonstrated that the technical ability of force application is strongly associated with the extensional component of forward CoM velocity regardless of acceleration task (Table 2). A higher extensional contribution was positively associated with RF. As CoM extension can only translate the CoM forward for CoM-angles greater than 90° (Figure 2), both increasing the angle of attack and rapid CoM rotation after TD to move the CoM forward are beneficial for increasing the extensional velocity component.

The interdependence of technical ability, body position and CoM velocity components become clearer when contrasting unresisted and resisted acceleration. Accelerating an external mass produced a higher RF compared to the unresisted acceleration (.44 vs .28) and provoked an increase in the angle of attack (88.9° vs. 75.4°) and take-off angle (139.3° vs. 128.8°). Interestingly, the sector covered by CoM rotation during the ground contact phases remained unaffected, suggesting that the athletes did not rotate less during resisted acceleration, but shifted the sector for rotation forwards (Figure 3). The observed right-shift in this study for resisted acceleration increased the step-averaged propulsive effectiveness of CoM extension from .20 to .39. This increase was also reflected in significantly higher extensional contributions for both step-averaged values (29.5% vs. 17.2%) as well as for each individual ground contact phase (Table 3). Consequently, the technical ability to direct the GRF vector forwards while pushing an external mass coincides with a shift in the kinematic strategy from rotation towards extension. High shares in extensional velocity components have also previously been identified as a feature of superior accelerators during the bobsleigh push phase (Zedler et al., 2026). It should be noted that the rotational velocity component in resisted acceleration was still higher than the extensional component but increasing the extensional component constituted a kinematic feature to improve the technical ability of force application.

The observed changes in velocity components between acceleration tasks were retrospectively investigated using a log-difference decomposition of the underlying equations of rotational and extensional velocity components. The decrease in the rotational velocity component for the resisted task ($\Delta\log(v_{rot})$) was predominantly explained by a decrease in the angular velocity of CoM rotation ($\Delta\log(\dot{\alpha})$, 65.4%), followed by a decrease in the propulsive effectiveness of CoM rotation ($\Delta\log(\sin(\alpha))$, 21.9%) and a decrease in leg length ($\Delta\log(L)$, 12.7%). In contrast, the increase in the extensional component for the resisted task ($\Delta\log(v_{ext})$) was mainly driven by an increase in the propulsive effectiveness of CoM extension ($\Delta\log(\cos(\alpha))$, 64.2%) followed by an increase in CoM extension velocity ($\Delta\log(\dot{L})$, 35.8%). This decomposition suggests that increasing extensional contribution to improve technical ability could be achieved by adopting a greater whole-body forward orientation, lower CoM position to increase the distance for CoM extension, and explosive lower limb extension in late stance.

The shank segment emerged to play a key role in setting up a body configuration for effective force application to accelerate the CoM forwards. First, shank segment orientation had the greatest influence in modulating the angle of attack (Table 4) independent of acceleration task, and second, by its forward rotation during stance, the shank segment set up a kinetic chain to enable effective foot and thigh rotation. This dual role highlights the importance of distal segment configuration to produce propulsion, which is in line with previous studies (Bezodis et al., 2015; King et al., 2022).

As the angle of attack can only be manipulated prior to TD, the shank motion during the late flight phase preceding TD requires attention from practitioners and researchers. During the late flight phase, the shank of the leading leg was described to reverse its rotation direction from backward to forward (shin block event) and to progressively rotate forward in preparation of the subsequent ground contact (shin alignment strategy) (Alt et al., 2022). Reaching the shin block event earlier would allow more time to rotate the shank segment forward in preparation for TD, or reaching the shin block with a greater forward oriented shank could

both be strategies to increase shank forward orientation at TD and therefore the angle of attack. From an applied perspective, this suggest that technical interventions should focus on shin block timing by 'actively catching' the backward rotating shank of the leading leg. Since the reversal of shank rotation direction must be realised, while the hip is flexed and the knee joint extends (Alt et al., 2021), the biarticular parts of the hamstring muscles must withstand high eccentric forces. Therefore, these mechanical demands should be mimicked in sprint-specific strength and conditioning exercises.

Previous studies on resisted acceleration have shown that the ankle joint dorsiflexed after TD and then remained dorsiflexed until plantarflexion and positive mechanical work production at the ankle joint were initiated in late stance (Zedler et al., 2025)—a pattern similarly observed for unresisted acceleration (Schache et al., 2019). The present findings further elaborate this sequence, by investigating segmental rotations and their effect on propulsion from a geometrical perspective. During the period of sustained dorsiflexion angle, the foot and shank continuously rotated forward, and when the shank reached its greatest forward orientation, a distinct increase in foot rotation—induced by the onset of plantarflexion—was observed (Figure 5 for resisted acceleration). Delaying plantarflexion until sufficient forward shank rotation has been achieved may enhance the propulsive effectiveness of foot rotation, whereas initiating plantarflexion prematurely, when the shank is still relatively vertical, would be mechanically less effective. These findings further support the principle of posture before power and add a segment-level explanation of which was termed propulsion pose (Alt et al., 2022). For practitioners, this implies that coaching cues should prioritise (i) early preparation of the shank during late flight, (ii) patience in delaying plantarflexion until late stance, and (iii) maintaining a relatively low CoM position to improve effective CoM extension. Such cues are likely applicable across sprint acceleration, resisted sprint training, and the bobsleigh push phase, where effective forward orientation and coordinated segmental timing are critical for horizontal force production.

The participants in this study consisted of a highly trained elite cohort of male bobsleigh athletes. Accordingly, caution is advised when transferring practical applications to different populations. Implementing different acceleration tasks in the analyses aimed at highlighting relationships between mechanical effectiveness of force application and kinematics as well as providing a high specificity of the findings for bobsleigh. Resisted acceleration was therefore realised via sled-pushing rather than pulling, with a sled mass chosen to approximate the mechanical demands during the bobsleigh push phase (Zedler et al., 2025). While this enhances sport-specific relevance, it may limit generalisability to other resisted sprint modalities that differ in posture constraints (i.e., sled-towing). The fixed sequencing of experimental conditions (unresisted preceding resisted acceleration) was chosen for practical reasons and may have introduced order-related effects despite sufficient rest periods between trials. Only the fastest trial per condition was analysed to examine kinematic strategies underpinning effective force application under maximal effort. While this approach limits insight into within-athlete variability, the elite status of the cohort suggests limited inter-trial performance variability. Finally, potential relationships between segmental rotations and joint kinetics were beyond the scope of this study and represent an interesting avenue for future research.

Conclusion

This study advanced the concept of technical ability of force application in sprint acceleration on a kinematic basis. Technical ability was strongly associated with increased CoM and lower limb segment forward orientation, and a shift of the kinematic strategy to produce propulsion in favour of CoM extension for both unresisted and resisted acceleration. The shank emerged as the key segment to modulate body position and to set up a configuration of the lower limb segment chain beneficial for anterior GRF production. Fast feedback tools should therefore focus on CoM-angle and shank rotation of the swing leg preceding TD and the subsequent stance phase.

Author contributions

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